

$$3d^5$$

August 13, 2026

$$1 \quad \left| 3d_{x^2-y^2}^2 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{z^2}^\alpha 3d_{xy}^0 \right\rangle$$

Here are the CSFs:

$$\begin{aligned}
|1, 5/2, 5/2\rangle &= \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \\
|2, 5/2, 3/2\rangle &= \frac{1}{\sqrt{5}} \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{5}} \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
&\quad + \frac{1}{\sqrt{5}} \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle + \frac{1}{\sqrt{5}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \\
&\quad + \frac{1}{\sqrt{5}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \\
|3, 5/2, 1/2\rangle &= \frac{1}{\sqrt{10}} \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{10}} \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad + \frac{1}{\sqrt{10}} \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle + \frac{1}{\sqrt{10}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad + \frac{1}{\sqrt{10}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle + \frac{1}{\sqrt{10}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \\
&\quad + \frac{1}{\sqrt{10}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{10}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \\
&\quad + \frac{1}{\sqrt{10}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle + \frac{1}{\sqrt{10}} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle
\end{aligned}$$

$$\begin{aligned}
|12, 3/2, 1/2\rangle &= +\frac{2}{3\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle + \frac{2}{3\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
&\quad - \frac{1}{3\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{3\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
&\quad + \frac{1}{3\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle - \frac{1}{3\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad - \frac{1}{3\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle + \frac{1}{3\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
&\quad - \frac{2}{3\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \\
|13, 3/2, -1/2\rangle &= \frac{2}{3\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{3\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad + \frac{1}{3\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{3\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
&\quad - \frac{1}{3\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{3\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
&\quad + \frac{1}{3\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle - \frac{2}{3\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad - \frac{2}{3\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
|14, 3/2, -3/2\rangle &= \frac{1}{\sqrt{6}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{6}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad - \sqrt{\frac{2}{3}} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle \\
|15, 3/2, 3/2\rangle &= +\frac{\sqrt{3}}{2} \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle - \frac{1}{\sqrt{12}} \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \\
&\quad - \frac{1}{\sqrt{12}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle - \frac{1}{\sqrt{12}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \\
|16, 3/2, 1/2\rangle &= \frac{1}{2} \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{6} \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad + \frac{1}{3} \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle - \frac{1}{6} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad + \frac{1}{3} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle - \frac{1}{3} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \\
&\quad - \frac{1}{6} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{3} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \\
&\quad - \frac{1}{3} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle - \frac{1}{3} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle
\end{aligned}$$

$$\begin{aligned}
|17, 3/2, -1/2\rangle &= \frac{1}{3} \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{3} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad - \frac{1}{3} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{6} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
&\quad + \frac{1}{3} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{3} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad + \frac{1}{6} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle - \frac{1}{3} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad + \frac{1}{6} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle - \frac{1}{2} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
|18, 3/2, -3/2\rangle &= \frac{1}{\sqrt{12}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{12}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad + \frac{1}{\sqrt{12}} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle - \frac{\sqrt{3}}{2} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle \\
|19, 3/2, 3/2\rangle &= \frac{2}{\sqrt{5}} \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{2\sqrt{5}} \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
&\quad - \frac{1}{2\sqrt{5}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{2\sqrt{5}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \\
&\quad - \frac{1}{2\sqrt{5}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \\
|20, 3/2, 1/2\rangle &= \frac{\sqrt{3}}{2\sqrt{5}} \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle + \frac{\sqrt{3}}{2\sqrt{5}} \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
&\quad - \frac{\sqrt{3}}{3\sqrt{5}} \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle + \frac{\sqrt{3}}{2\sqrt{5}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad - \frac{\sqrt{3}}{3\sqrt{5}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle - \frac{\sqrt{3}}{3\sqrt{5}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad + \frac{\sqrt{3}}{2\sqrt{5}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle - \frac{\sqrt{3}}{3\sqrt{5}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
&\quad - \frac{\sqrt{3}}{3\sqrt{5}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle - \frac{\sqrt{3}}{3\sqrt{5}} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle
\end{aligned}$$

$$\begin{aligned}
|21, 3/2, -1/2\rangle &= \frac{\sqrt{3}}{3\sqrt{5}} \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle + \frac{\sqrt{3}}{3\sqrt{5}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle \\
&+ \frac{\sqrt{3}}{3\sqrt{5}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle - \frac{\sqrt{3}}{2\sqrt{5}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
&+ \frac{\sqrt{3}}{3\sqrt{5}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle + \frac{\sqrt{3}}{3\sqrt{5}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle \\
&- \frac{\sqrt{3}}{2\sqrt{5}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle + \frac{\sqrt{3}}{3\sqrt{5}} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle \\
&- \frac{\sqrt{3}}{2\sqrt{5}} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle - \frac{\sqrt{3}}{2\sqrt{5}} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle \\
|22, 3/2, -3/2\rangle &= \frac{1}{2\sqrt{5}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{2\sqrt{5}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle \\
&+ \frac{1}{2\sqrt{5}} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{2\sqrt{5}} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle \\
&- \frac{2}{\sqrt{5}} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
|23, 3/2, 3/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^0 \right\rangle \\
|24, 3/2, 3/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^\alpha \right\rangle \\
|25, 3/2, 3/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^0 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \\
|26, 3/2, 3/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^0 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \\
|27, 3/2, 3/2\rangle &= \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^0 \right\rangle \\
|28, 3/2, 3/2\rangle &= \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^\alpha \right\rangle \\
|29, 3/2, 3/2\rangle &= \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \\
|30, 3/2, 3/2\rangle &= \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^0 \right\rangle \\
|31, 3/2, 3/2\rangle &= \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^\alpha \right\rangle \\
|32, 3/2, 3/2\rangle &= \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^0 \right\rangle \\
|33, 3/2, 3/2\rangle &= \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^2 \right\rangle \\
|34, 3/2, 3/2\rangle &= \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle
\end{aligned}$$

$$\begin{aligned}
|35, 3/2, 3/2\rangle &= \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^0 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|36, 3/2, 3/2\rangle &= \left| 3d_{xz}^\alpha 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \\
|37, 3/2, 3/2\rangle &= \left| 3d_{xz}^\alpha 3d_{yz}^0 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|38, 3/2, 3/2\rangle &= \left| 3d_{xz}^\alpha 3d_{yz}^0 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|39, 3/2, 3/2\rangle &= \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \\
|40, 3/2, 3/2\rangle &= \left| 3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \\
|41, 3/2, 3/2\rangle &= \left| 3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|42, 3/2, 3/2\rangle &= \left| 3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|43, 3/2, 1/2\rangle &= \frac{1}{\sqrt{3}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^0 \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^0 \right\rangle \\
&\quad + \frac{1}{\sqrt{3}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^0 \right\rangle \\
|44, 3/2, 1/2\rangle &= \frac{1}{\sqrt{3}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^0 3d_{x^2-y^2}^\alpha \right\rangle \\
&\quad + \frac{1}{\sqrt{3}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^\alpha \right\rangle \\
|45, 3/2, 1/2\rangle &= \frac{1}{\sqrt{3}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^0 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^0 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
&\quad + \frac{1}{\sqrt{3}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \\
|46, 3/2, 1/2\rangle &= \frac{1}{\sqrt{3}} \left| 3d_{xz}^2 3d_{yz}^0 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^2 3d_{yz}^0 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
&\quad + \frac{1}{\sqrt{3}} \left| 3d_{xz}^2 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \\
|47, 3/2, 1/2\rangle &= \frac{1}{\sqrt{3}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^0 \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^0 \right\rangle \\
&\quad + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^0 \right\rangle \\
|48, 3/2, 1/2\rangle &= \frac{1}{\sqrt{3}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^0 3d_{x^2-y^2}^\alpha \right\rangle \\
&\quad + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^\alpha \right\rangle
\end{aligned}$$

$$\begin{aligned}
|49, 3/2, 1/2\rangle &= \frac{1}{\sqrt{3}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
&\quad + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \\
|50, 3/2, 1/2\rangle &= \frac{1}{\sqrt{3}} \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^0 \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^0 \right\rangle \\
&\quad + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^0 \right\rangle \\
|51, 3/2, 1/2\rangle &= \frac{1}{\sqrt{3}} \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^\alpha \right\rangle \\
&\quad + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^\alpha \right\rangle \\
|52, 3/2, 1/2\rangle &= \frac{1}{\sqrt{3}} \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^0 \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^0 \right\rangle \\
&\quad + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^0 \right\rangle \\
|53, 3/2, 1/2\rangle &= \frac{1}{\sqrt{3}} \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^0 3d_{x^2-y^2}^2 \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^2 \right\rangle \\
&\quad + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^2 \right\rangle \\
|54, 3/2, 1/2\rangle &= \frac{1}{\sqrt{3}} \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
&\quad + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|55, 3/2, 1/2\rangle &= \frac{1}{\sqrt{3}} \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^0 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
&\quad + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^0 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|56, 3/2, 1/2\rangle &= \frac{1}{\sqrt{3}} \left| 3d_{xz}^\alpha 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\alpha 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
&\quad + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \\
|57, 3/2, 1/2\rangle &= \frac{1}{\sqrt{3}} \left| 3d_{xz}^\alpha 3d_{yz}^0 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\alpha 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
&\quad + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle
\end{aligned}$$

$$\begin{aligned}
|58, 3/2, 1/2\rangle &= \frac{1}{\sqrt{3}} \left| 3d_{xz}^\alpha 3d_{yz}^0 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\alpha 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
&\quad + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|59, 3/2, 1/2\rangle &= \frac{1}{\sqrt{3}} \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
&\quad + \frac{1}{\sqrt{3}} \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \\
|60, 3/2, 1/2\rangle &= \frac{1}{\sqrt{3}} \left| 3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
&\quad + \frac{1}{\sqrt{3}} \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \\
|61, 3/2, 1/2\rangle &= \frac{1}{\sqrt{3}} \left| 3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
&\quad + \frac{1}{\sqrt{3}} \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|62, 3/2, 1/2\rangle &= \frac{1}{\sqrt{3}} \left| 3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
&\quad + \frac{1}{\sqrt{3}} \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|63, 3/2, -1/2\rangle &= \frac{1}{\sqrt{3}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^0 \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^0 \right\rangle \\
&\quad + \frac{1}{\sqrt{3}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^0 \right\rangle \\
|64, 3/2, -1/2\rangle &= \frac{1}{\sqrt{3}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^0 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad + \frac{1}{\sqrt{3}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^0 3d_{x^2-y^2}^\alpha \right\rangle \\
|65, 3/2, -1/2\rangle &= \frac{1}{\sqrt{3}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^0 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad + \frac{1}{\sqrt{3}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
|66, 3/2, -1/2\rangle &= \frac{1}{\sqrt{3}} \left| 3d_{xz}^2 3d_{yz}^0 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^2 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad + \frac{1}{\sqrt{3}} \left| 3d_{xz}^2 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle
\end{aligned}$$

$$\begin{aligned}
|67, 3/2, -1/2\rangle &= \frac{1}{\sqrt{3}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^0 \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^0 \right\rangle \\
&\quad + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^0 \right\rangle \\
|68, 3/2, -1/2\rangle &= \frac{1}{\sqrt{3}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^0 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^0 3d_{x^2-y^2}^\alpha \right\rangle \\
|69, 3/2, -1/2\rangle &= \frac{1}{\sqrt{3}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
|70, 3/2, -1/2\rangle &= \frac{1}{\sqrt{3}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^0 \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^0 \right\rangle \\
&\quad + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^0 \right\rangle \\
|71, 3/2, -1/2\rangle &= \frac{1}{\sqrt{3}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^\alpha \right\rangle \\
|72, 3/2, -1/2\rangle &= \frac{1}{\sqrt{3}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^0 \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^0 \right\rangle \\
&\quad + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^0 \right\rangle \\
|73, 3/2, -1/2\rangle &= \frac{1}{\sqrt{3}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^0 3d_{x^2-y^2}^2 \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^0 3d_{x^2-y^2}^2 \right\rangle \\
&\quad + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^2 \right\rangle \\
|74, 3/2, -1/2\rangle &= \frac{1}{\sqrt{3}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|75, 3/2, -1/2\rangle &= \frac{1}{\sqrt{3}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^0 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle \\
&\quad + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle
\end{aligned}$$

$$\begin{aligned}
|76, 3/2, -1/2\rangle &= \frac{1}{\sqrt{3}} \left| 3d_{xz}^\alpha 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
|77, 3/2, -1/2\rangle &= \frac{1}{\sqrt{3}} \left| 3d_{xz}^\alpha 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|78, 3/2, -1/2\rangle &= \frac{1}{\sqrt{3}} \left| 3d_{xz}^\alpha 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle \\
&\quad + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|79, 3/2, -1/2\rangle &= \frac{1}{\sqrt{3}} \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad + \frac{1}{\sqrt{3}} \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
|80, 3/2, -1/2\rangle &= \frac{1}{\sqrt{3}} \left| 3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad + \frac{1}{\sqrt{3}} \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
|81, 3/2, -1/2\rangle &= \frac{1}{\sqrt{3}} \left| 3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad + \frac{1}{\sqrt{3}} \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|82, 3/2, -1/2\rangle &= \frac{1}{\sqrt{3}} \left| 3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle \\
&\quad + \frac{1}{\sqrt{3}} \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|83, 3/2, -3/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^0 \right\rangle \\
|84, 3/2, -3/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^0 3d_{x^2-y^2}^\beta \right\rangle \\
|85, 3/2, -3/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle \\
|86, 3/2, -3/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle \\
|87, 3/2, -3/2\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^0 \right\rangle
\end{aligned}$$

$$\begin{aligned}
|88, 3/2, -3/2\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^0 3d_{x^2-y^2}^\beta \right\rangle \\
|89, 3/2, -3/2\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle \\
|90, 3/2, -3/2\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^0 \right\rangle \\
|91, 3/2, -3/2\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^\beta \right\rangle \\
|92, 3/2, -3/2\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^0 \right\rangle \\
|93, 3/2, -3/2\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^0 3d_{x^2-y^2}^2 \right\rangle \\
|94, 3/2, -3/2\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle \\
|95, 3/2, -3/2\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle \\
|96, 3/2, -3/2\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle \\
|97, 3/2, -3/2\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle \\
|98, 3/2, -3/2\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle \\
|99, 3/2, -3/2\rangle &= \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle \\
|100, 3/2, -3/2\rangle &= \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle \\
|101, 3/2, -3/2\rangle &= \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle \\
|102, 3/2, -3/2\rangle &= \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle \\
|103, 1/2, 1/2\rangle &= +\frac{1}{2} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle - \frac{1}{2} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \\
&\quad - \frac{1}{2} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle + \frac{1}{2} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \\
|104, 1/2, -1/2\rangle &= +\frac{1}{2} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{2} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad - \frac{1}{2} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{2} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle
\end{aligned}$$

[illegible]

[illegible]

$$\begin{aligned}
|153, 1/2, 1/2\rangle &= \sqrt{\frac{2}{3}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^0 \right\rangle - \frac{1}{\sqrt{6}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^0 \right\rangle \\
&\quad - \frac{1}{\sqrt{6}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^0 \right\rangle \\
|154, 1/2, 1/2\rangle &= \sqrt{\frac{2}{3}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{\sqrt{6}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^0 3d_{x^2-y^2}^\alpha \right\rangle \\
&\quad - \frac{1}{\sqrt{6}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^\alpha \right\rangle \\
|155, 1/2, 1/2\rangle &= \sqrt{\frac{2}{3}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^0 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{\sqrt{6}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^0 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
&\quad - \frac{1}{\sqrt{6}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \\
|156, 1/2, 1/2\rangle &= \sqrt{\frac{2}{3}} \left| 3d_{xz}^2 3d_{yz}^0 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{\sqrt{6}} \left| 3d_{xz}^2 3d_{yz}^0 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
&\quad - \frac{1}{\sqrt{6}} \left| 3d_{xz}^2 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \\
|157, 1/2, 1/2\rangle &= \sqrt{\frac{2}{3}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^0 \right\rangle - \frac{1}{\sqrt{6}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^0 \right\rangle \\
&\quad - \frac{1}{\sqrt{6}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^0 \right\rangle \\
|158, 1/2, 1/2\rangle &= \sqrt{\frac{2}{3}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{\sqrt{6}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^0 3d_{x^2-y^2}^\alpha \right\rangle \\
&\quad - \frac{1}{\sqrt{6}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^\alpha \right\rangle \\
|159, 1/2, 1/2\rangle &= \sqrt{\frac{2}{3}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{\sqrt{6}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
&\quad - \frac{1}{\sqrt{6}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \\
|160, 1/2, 1/2\rangle &= \sqrt{\frac{2}{3}} \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^0 \right\rangle - \frac{1}{\sqrt{6}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^0 \right\rangle \\
&\quad - \frac{1}{\sqrt{6}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^0 \right\rangle \\
|161, 1/2, 1/2\rangle &= \sqrt{\frac{2}{3}} \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{\sqrt{6}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^\alpha \right\rangle \\
&\quad - \frac{1}{\sqrt{6}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^\alpha \right\rangle
\end{aligned}$$

$$\begin{aligned}
|162, 1/2, 1/2\rangle &= \sqrt{\frac{2}{3}} \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^0 \right\rangle - \frac{1}{\sqrt{6}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^0 \right\rangle \\
&\quad - \frac{1}{\sqrt{6}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^0 \right\rangle \\
|163, 1/2, 1/2\rangle &= \sqrt{\frac{2}{3}} \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^0 3d_{x^2-y^2}^2 \right\rangle - \frac{1}{\sqrt{6}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^2 \right\rangle \\
&\quad - \frac{1}{\sqrt{6}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^2 \right\rangle \\
|164, 1/2, 1/2\rangle &= \sqrt{\frac{2}{3}} \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{\sqrt{6}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
&\quad - \frac{1}{\sqrt{6}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|165, 1/2, 1/2\rangle &= \sqrt{\frac{2}{3}} \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^0 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle - \frac{1}{\sqrt{6}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
&\quad - \frac{1}{\sqrt{6}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^0 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|166, 1/2, 1/2\rangle &= \sqrt{\frac{2}{3}} \left| 3d_{xz}^\alpha 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{\sqrt{6}} \left| 3d_{xz}^\alpha 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
&\quad - \frac{1}{\sqrt{6}} \left| 3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \\
|167, 1/2, 1/2\rangle &= \sqrt{\frac{2}{3}} \left| 3d_{xz}^\alpha 3d_{yz}^0 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{\sqrt{6}} \left| 3d_{xz}^\alpha 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
&\quad - \frac{1}{\sqrt{6}} \left| 3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|168, 1/2, 1/2\rangle &= \sqrt{\frac{2}{3}} \left| 3d_{xz}^\alpha 3d_{yz}^0 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle - \frac{1}{\sqrt{6}} \left| 3d_{xz}^\alpha 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
&\quad - \frac{1}{\sqrt{6}} \left| 3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|169, 1/2, 1/2\rangle &= \sqrt{\frac{2}{3}} \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{\sqrt{6}} \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
&\quad - \frac{1}{\sqrt{6}} \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \\
|170, 1/2, 1/2\rangle &= \sqrt{\frac{2}{3}} \left| 3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{\sqrt{6}} \left| 3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
&\quad - \frac{1}{\sqrt{6}} \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle
\end{aligned}$$

$$\begin{aligned}
|171, 1/2, 1/2\rangle &= \sqrt{\frac{2}{3}} \left| 3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{\sqrt{6}} \left| 3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
&\quad - \frac{1}{\sqrt{6}} \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|172, 1/2, 1/2\rangle &= \sqrt{\frac{2}{3}} \left| 3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle - \frac{1}{\sqrt{6}} \left| 3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
&\quad - \frac{1}{\sqrt{6}} \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|173, 1/2, -1/2\rangle &= \frac{1}{\sqrt{6}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^0 \right\rangle + \frac{1}{\sqrt{6}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^0 \right\rangle \\
&\quad - \sqrt{\frac{2}{3}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^0 \right\rangle \\
|174, 1/2, -1/2\rangle &= \frac{1}{\sqrt{6}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^0 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{6}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad - \sqrt{\frac{2}{3}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^0 3d_{x^2-y^2}^\alpha \right\rangle \\
|175, 1/2, -1/2\rangle &= \frac{1}{\sqrt{6}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^0 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{6}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad - \sqrt{\frac{2}{3}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
|176, 1/2, -1/2\rangle &= \frac{1}{\sqrt{6}} \left| 3d_{xz}^2 3d_{yz}^0 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{6}} \left| 3d_{xz}^2 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad - \sqrt{\frac{2}{3}} \left| 3d_{xz}^2 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
|177, 1/2, -1/2\rangle &= \frac{1}{\sqrt{6}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^0 \right\rangle + \frac{1}{\sqrt{6}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^0 \right\rangle \\
&\quad - \sqrt{\frac{2}{3}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^0 \right\rangle \\
|178, 1/2, -1/2\rangle &= \frac{1}{\sqrt{6}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^0 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{6}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad - \sqrt{\frac{2}{3}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^0 3d_{x^2-y^2}^\alpha \right\rangle \\
|179, 1/2, -1/2\rangle &= \frac{1}{\sqrt{6}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{6}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad - \sqrt{\frac{2}{3}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle
\end{aligned}$$

$$\begin{aligned}
|180, 1/2, -1/2\rangle &= \frac{1}{\sqrt{6}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^0 \right\rangle + \frac{1}{\sqrt{6}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^0 \right\rangle \\
&\quad - \sqrt{\frac{2}{3}} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^0 \right\rangle \\
|181, 1/2, -1/2\rangle &= \frac{1}{\sqrt{6}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{6}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad - \sqrt{\frac{2}{3}} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^\alpha \right\rangle \\
|182, 1/2, -1/2\rangle &= \frac{1}{\sqrt{6}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^0 \right\rangle + \frac{1}{\sqrt{6}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^0 \right\rangle \\
&\quad - \sqrt{\frac{2}{3}} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^0 \right\rangle \\
|183, 1/2, -1/2\rangle &= \frac{1}{\sqrt{6}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^0 3d_{x^2-y^2}^2 \right\rangle + \frac{1}{\sqrt{6}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^0 3d_{x^2-y^2}^2 \right\rangle \\
&\quad - \sqrt{\frac{2}{3}} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^2 \right\rangle \\
|184, 1/2, -1/2\rangle &= \frac{1}{\sqrt{6}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{6}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad - \sqrt{\frac{2}{3}} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|185, 1/2, -1/2\rangle &= \frac{1}{\sqrt{6}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle + \frac{1}{\sqrt{6}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^0 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle \\
&\quad - \sqrt{\frac{2}{3}} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|186, 1/2, -1/2\rangle &= \frac{1}{\sqrt{6}} \left| 3d_{xz}^\alpha 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{6}} \left| 3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad - \sqrt{\frac{2}{3}} \left| 3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
|187, 1/2, -1/2\rangle &= \frac{1}{\sqrt{6}} \left| 3d_{xz}^\alpha 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{6}} \left| 3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad - \sqrt{\frac{2}{3}} \left| 3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|188, 1/2, -1/2\rangle &= \frac{1}{\sqrt{6}} \left| 3d_{xz}^\alpha 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle + \frac{1}{\sqrt{6}} \left| 3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle \\
&\quad - \sqrt{\frac{2}{3}} \left| 3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle
\end{aligned}$$

$$\begin{aligned}
|189, 1/2, -1/2\rangle &= \frac{1}{\sqrt{6}} \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{6}} \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad - \sqrt{\frac{2}{3}} \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
|190, 1/2, -1/2\rangle &= \frac{1}{\sqrt{6}} \left| 3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{6}} \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad - \sqrt{\frac{2}{3}} \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
|191, 1/2, -1/2\rangle &= \frac{1}{\sqrt{6}} \left| 3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{6}} \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad - \sqrt{\frac{2}{3}} \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|192, 1/2, -1/2\rangle &= \frac{1}{\sqrt{6}} \left| 3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle + \frac{1}{\sqrt{6}} \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle \\
&\quad - \sqrt{\frac{2}{3}} \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|193, 1/2, 1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^0 \right\rangle \\
|194, 1/2, 1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^\alpha 3d_{x^2-y^2}^0 \right\rangle \\
|195, 1/2, 1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^0 3d_{x^2-y^2}^\alpha \right\rangle \\
|196, 1/2, 1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^0 \right\rangle \\
|197, 1/2, 1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^0 \right\rangle \\
|198, 1/2, 1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^0 3d_{z^2}^0 3d_{x^2-y^2}^2 \right\rangle \\
|199, 1/2, 1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^0 \right\rangle \\
|200, 1/2, 1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^\alpha \right\rangle \\
|201, 1/2, 1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^0 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^0 \right\rangle \\
|202, 1/2, 1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^0 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^2 \right\rangle \\
|203, 1/2, 1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^0 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|204, 1/2, 1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^0 3d_{xy}^0 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle
\end{aligned}$$

$$\begin{aligned}
|205, 1/2, 1/2\rangle &= \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^0 \right\rangle \\
|206, 1/2, 1/2\rangle &= \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^0 \right\rangle \\
|207, 1/2, 1/2\rangle &= \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^0 3d_{x^2-y^2}^2 \right\rangle \\
|208, 1/2, 1/2\rangle &= \left| 3d_{xz}^\alpha 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^0 \right\rangle \\
|209, 1/2, 1/2\rangle &= \left| 3d_{xz}^\alpha 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^2 \right\rangle \\
|210, 1/2, 1/2\rangle &= \left| 3d_{xz}^\alpha 3d_{yz}^0 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|211, 1/2, 1/2\rangle &= \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^0 \right\rangle \\
|212, 1/2, 1/2\rangle &= \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^\alpha \right\rangle \\
|213, 1/2, 1/2\rangle &= \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^0 \right\rangle \\
|214, 1/2, 1/2\rangle &= \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^2 \right\rangle \\
|215, 1/2, 1/2\rangle &= \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|216, 1/2, 1/2\rangle &= \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|217, 1/2, 1/2\rangle &= \left| 3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^0 \right\rangle \\
|218, 1/2, 1/2\rangle &= \left| 3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^2 \right\rangle \\
|219, 1/2, 1/2\rangle &= \left| 3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|220, 1/2, 1/2\rangle &= \left| 3d_{xz}^0 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|221, 1/2, 1/2\rangle &= \left| 3d_{xz}^0 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|222, 1/2, 1/2\rangle &= \left| 3d_{xz}^0 3d_{yz}^0 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|223, 1/2, -1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^0 3d_{x^2-y^2}^0 \right\rangle \\
|224, 1/2, -1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^\beta 3d_{x^2-y^2}^0 \right\rangle \\
|225, 1/2, -1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^0 3d_{x^2-y^2}^\beta \right\rangle
\end{aligned}$$

$$\begin{aligned}
|226, 1/2, -1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^0 \right\rangle \\
|227, 1/2, -1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^0 \right\rangle \\
|228, 1/2, -1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^0 3d_{x^2-y^2}^2 \right\rangle \\
|229, 1/2, -1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^0 \right\rangle \\
|230, 1/2, -1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^\beta \right\rangle \\
|231, 1/2, -1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^0 \right\rangle \\
|232, 1/2, -1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^0 3d_{x^2-y^2}^2 \right\rangle \\
|233, 1/2, -1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^0 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle \\
|234, 1/2, -1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^0 3d_{xy}^0 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle \\
|235, 1/2, -1/2\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^0 \right\rangle \\
|236, 1/2, -1/2\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^0 \right\rangle \\
|237, 1/2, -1/2\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^0 3d_{x^2-y^2}^2 \right\rangle \\
|238, 1/2, -1/2\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^0 \right\rangle \\
|239, 1/2, -1/2\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^2 \right\rangle \\
|240, 1/2, -1/2\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|241, 1/2, -1/2\rangle &= \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^0 \right\rangle \\
|242, 1/2, -1/2\rangle &= \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^\beta \right\rangle \\
|243, 1/2, -1/2\rangle &= \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^0 \right\rangle \\
|244, 1/2, -1/2\rangle &= \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^0 3d_{x^2-y^2}^2 \right\rangle \\
|245, 1/2, -1/2\rangle &= \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle \\
|246, 1/2, -1/2\rangle &= \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle
\end{aligned}$$

$$\begin{aligned}
|247, 1/2, -1/2\rangle &= \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^0 \right\rangle \\
|248, 1/2, -1/2\rangle &= \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^2 \right\rangle \\
|249, 1/2, -1/2\rangle &= \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|250, 1/2, -1/2\rangle &= \left| 3d_{xz}^0 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle \\
|251, 1/2, -1/2\rangle &= \left| 3d_{xz}^0 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle \\
|252, 1/2, -1/2\rangle &= \left| 3d_{xz}^0 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle
\end{aligned}$$

The Ground State CSFs are 35 55 75 95

$$\begin{aligned}
\left\langle 35, \frac{3}{2}, \frac{3}{2} \left| L_x \right| 29, \frac{3}{2}, \frac{3}{2} \right\rangle &= I \\
\left\langle 35, \frac{3}{2}, \frac{3}{2} \left| L_x \right| 42, \frac{3}{2}, \frac{3}{2} \right\rangle &= -I
\end{aligned}$$

Δg_{xx}

$$\begin{aligned}
&-\frac{2}{3}\zeta\Delta_{35-29}^{-1} \\
&-\frac{2}{3}\zeta\Delta_{35-42}^{-1}
\end{aligned}$$

$$\begin{aligned}
\left\langle 35, \frac{3}{2}, \frac{3}{2} \left| L_y \right| 25, \frac{3}{2}, \frac{3}{2} \right\rangle &= -I \\
\left\langle 35, \frac{3}{2}, \frac{3}{2} \left| L_y \right| 38, \frac{3}{2}, \frac{3}{2} \right\rangle &= -I
\end{aligned}$$

Δg_{yy}

$$\begin{aligned}
&-\frac{2}{3}\zeta\Delta_{35-25}^{-1} \\
&-\frac{2}{3}\zeta\Delta_{35-38}^{-1}
\end{aligned}$$

$$\begin{aligned}
\left\langle 35, \frac{3}{2}, \frac{3}{2} \left| L_z \right| 11, \frac{3}{2}, \frac{3}{2} \right\rangle &= -2\sqrt{\frac{2}{3}}I \\
\left\langle 35, \frac{3}{2}, \frac{3}{2} \left| L_z \right| 15, \frac{3}{2}, \frac{3}{2} \right\rangle &= \frac{1}{\sqrt{3}}I \\
\left\langle 35, \frac{3}{2}, \frac{3}{2} \left| L_z \right| 19, \frac{3}{2}, \frac{3}{2} \right\rangle &= \sqrt{5}I
\end{aligned}$$

Δg_{zz}

$$\begin{aligned}
&-\frac{16}{9}\zeta\Delta_{35-11}^{-1} \\
&-\frac{2}{9}\zeta\Delta_{35-15}^{-1} \\
&-\frac{10}{3}\zeta\Delta_{35-19}^{-1}
\end{aligned}$$

For a multiplicity of 4 we have the following:

$$D = \frac{1}{2} (\langle \frac{3}{2}, \frac{3}{2} | \frac{3}{2}, \frac{3}{2} \rangle - \langle \frac{3}{2}, \frac{1}{2} | \frac{3}{2}, \frac{1}{2} \rangle)$$

$$E = \frac{1}{\sqrt{3}} \langle \frac{3}{2}, -\frac{1}{2} | \frac{3}{2}, \frac{3}{2} \rangle$$

$$\begin{aligned} \left\langle \frac{3}{2}, \frac{3}{2} \left| \hat{H}_{eff} \right| \frac{3}{2}, \frac{3}{2} \right\rangle = & -\frac{4}{9}\zeta^2 (\\ & + \frac{4}{5}\Delta_2^{-1} + \frac{2}{3}\Delta_{11}^{-1} + \frac{1}{12}\Delta_{15}^{-1} + \frac{9}{20}\Delta_{19}^{-1} + \frac{1}{12}\Delta_{45}^{-1} + \frac{1}{12}\Delta_{49}^{-1} \\ & + \frac{1}{12}\Delta_{58}^{-1} + \frac{1}{12}\Delta_{62}^{-1} + \frac{1}{8}\Delta_{128}^{-1} + \frac{1}{8}\Delta_{132}^{-1} + \frac{1}{6}\Delta_{155}^{-1} + \frac{1}{6}\Delta_{159}^{-1} \\ & + \frac{1}{24}\Delta_{168}^{-1} + \frac{1}{24}\Delta_{172}^{-1} + \frac{3}{4}\Delta_{198}^{-1} + \frac{3}{4}\Delta_{207}^{-1} + \frac{3}{4}\Delta_{210}^{-1} + \frac{3}{4}\Delta_{219}^{-1} \\ &) \end{aligned}$$

$$\begin{aligned} \left\langle \frac{3}{2}, \frac{1}{2} \left| \hat{H}_{eff} \right| \frac{3}{2}, \frac{1}{2} \right\rangle = & -\frac{4}{9}\zeta^2 (\\ & + \frac{6}{5}\Delta_3^{-1} + \frac{2}{27}\Delta_{12}^{-1} + \frac{5}{54}\Delta_{16}^{-1} + \frac{1}{20}\Delta_{20}^{-1} + \frac{1}{12}\Delta_{25}^{-1} + \frac{1}{12}\Delta_{29}^{-1} \\ & + \frac{1}{12}\Delta_{38}^{-1} + \frac{1}{12}\Delta_{42}^{-1} + \frac{1}{9}\Delta_{65}^{-1} + \frac{1}{9}\Delta_{69}^{-1} + \frac{1}{9}\Delta_{78}^{-1} + \frac{1}{9}\Delta_{82}^{-1} \\ & + \frac{16}{27}\Delta_{109}^{-1} + \frac{2}{27}\Delta_{111}^{-1} + \frac{1}{24}\Delta_{148}^{-1} + \frac{1}{24}\Delta_{152}^{-1} + \frac{1}{18}\Delta_{175}^{-1} + \frac{1}{18}\Delta_{179}^{-1} \\ & + \frac{1}{72}\Delta_{188}^{-1} + \frac{1}{72}\Delta_{192}^{-1} + \frac{1}{3}\Delta_{204}^{-1} + \frac{1}{3}\Delta_{216}^{-1} + \frac{1}{4}\Delta_{228}^{-1} + \frac{1}{4}\Delta_{237}^{-1} \\ & + \frac{1}{4}\Delta_{240}^{-1} + \frac{1}{4}\Delta_{249}^{-1}) \end{aligned}$$

$$\begin{aligned} \left\langle \frac{3}{2}, -\frac{1}{2} \left| \hat{H}_{eff} \right| \frac{3}{2}, \frac{3}{2} \right\rangle = & -\frac{4}{9}\zeta^2 (\\ & - \frac{1}{3\sqrt{12}}\Delta_{45}^{-1} + \frac{1}{3\sqrt{12}}\Delta_{49}^{-1} - \frac{1}{3\sqrt{12}}\Delta_{58}^{-1} + \frac{1}{3\sqrt{12}}\Delta_{62}^{-1} + \frac{\sqrt{3}}{24}\Delta_{128}^{-1} \\ & - \frac{\sqrt{3}}{24}\Delta_{132}^{-1} + \frac{1}{3\sqrt{12}}\Delta_{155}^{-1} - \frac{1}{3\sqrt{12}}\Delta_{159}^{-1} + \frac{\sqrt{3}}{72}\Delta_{168}^{-1} - \frac{\sqrt{3}}{72}\Delta_{172}^{-1} \\ & + \frac{\sqrt{3}}{4}\Delta_{198}^{-1} - \frac{\sqrt{3}}{4}\Delta_{207}^{-1} - \frac{\sqrt{3}}{4}\Delta_{210}^{-1} + \frac{\sqrt{3}}{4}\Delta_{219}^{-1}) \end{aligned}$$

$$\begin{aligned}
D = & \frac{2}{9}\zeta^2 (\\
& -\frac{4}{5}\Delta_2^{-1} + \frac{6}{5}\Delta_3^{-1} - \frac{2}{3}\Delta_{11}^{-1} + \frac{2}{27}\Delta_{12}^{-1} - \frac{1}{12}\Delta_{15}^{-1} + \frac{5}{54}\Delta_{16}^{-1} \\
& -\frac{9}{20}\Delta_{19}^{-1} + \frac{1}{20}\Delta_{20}^{-1} + \frac{1}{12}\Delta_{25}^{-1} + \frac{1}{12}\Delta_{29}^{-1} + \frac{1}{12}\Delta_{38}^{-1} + \frac{1}{12}\Delta_{42}^{-1} \\
& -\frac{1}{12}\Delta_{45}^{-1} - \frac{1}{12}\Delta_{49}^{-1} - \frac{1}{12}\Delta_{58}^{-1} - \frac{1}{12}\Delta_{62}^{-1} + \frac{1}{9}\Delta_{65}^{-1} + \frac{1}{9}\Delta_{69}^{-1} \\
& +\frac{1}{9}\Delta_{78}^{-1} + \frac{1}{9}\Delta_{82}^{-1} + \frac{16}{27}\Delta_{109}^{-1} + \frac{2}{27}\Delta_{111}^{-1} - \frac{1}{8}\Delta_{128}^{-1} - \frac{1}{8}\Delta_{132}^{-1} \\
& +\frac{1}{24}\Delta_{148}^{-1} + \frac{1}{24}\Delta_{152}^{-1} - \frac{1}{6}\Delta_{155}^{-1} - \frac{1}{6}\Delta_{159}^{-1} - \frac{1}{24}\Delta_{168}^{-1} - \frac{1}{24}\Delta_{172}^{-1} \\
& +\frac{1}{18}\Delta_{175}^{-1} + \frac{1}{18}\Delta_{179}^{-1} + \frac{1}{72}\Delta_{188}^{-1} + \frac{1}{72}\Delta_{192}^{-1} - \frac{3}{4}\Delta_{198}^{-1} + \frac{1}{3}\Delta_{204}^{-1} \\
& -\frac{3}{4}\Delta_{207}^{-1} - \frac{3}{4}\Delta_{210}^{-1} + \frac{1}{3}\Delta_{216}^{-1} - \frac{3}{4}\Delta_{219}^{-1} + \frac{1}{4}\Delta_{228}^{-1} + \frac{1}{4}\Delta_{237}^{-1} \\
& +\frac{1}{4}\Delta_{240}^{-1} + \frac{1}{4}\Delta_{249}^{-1})
\end{aligned}$$

$$\begin{aligned}
E = & \frac{4}{9\sqrt{3}}\zeta^2 (\\
& +\frac{1}{3\sqrt{12}}\Delta_{45}^{-1} - \frac{1}{3\sqrt{12}}\Delta_{49}^{-1} + \frac{1}{3\sqrt{12}}\Delta_{58}^{-1} - \frac{1}{3\sqrt{12}}\Delta_{62}^{-1} - \frac{\sqrt{3}}{24}\Delta_{128}^{-1} \\
& +\frac{\sqrt{3}}{24}\Delta_{132}^{-1} - \frac{1}{3\sqrt{12}}\Delta_{155}^{-1} + \frac{1}{3\sqrt{12}}\Delta_{159}^{-1} - \frac{\sqrt{3}}{72}\Delta_{168}^{-1} + \frac{\sqrt{3}}{72}\Delta_{172}^{-1} \\
& -\frac{\sqrt{3}}{4}\Delta_{198}^{-1} + \frac{\sqrt{3}}{4}\Delta_{207}^{-1} + \frac{\sqrt{3}}{4}\Delta_{210}^{-1} - \frac{\sqrt{3}}{4}\Delta_{219}^{-1})
\end{aligned}$$